

HARSHA VARDHAN UPADRASTA

AI Automation QA Engineer | GenAI & Intelligent Quality Engineering | Azure OpenAI & Copilot Studio

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Automation QA Engineer with 4+ years of experience building intelligent quality-engineering systems across web, API, backend, and LLM-powered applications. Hands-on developer of Azure OpenAI and Copilot Studio agents, plus seven independently published open-source AI QA platforms spanning multi-agent test generation, self-healing locator intelligence, CI/CD quality gates, and automated root-cause analysis. Proficient in Python, Playwright, PyTest, TypeScript, and BDD, with a consistent record of embedding AI-driven quality gates into Azure DevOps and Jenkins pipelines to cut release-validation time and reduce defect leakage.

CORE SKILLS

Quality Engineering & Automation: Playwright, PyTest, BDD (Cucumber/Gherkin), E2E & Regression Testing, Integration Testing, Framework Architecture, Self-Healing Locators

Azure OpenAI & Copilot Studio: Agent Builder, Power Virtual Agents, Custom Connectors, Prompt Flow, Adaptive Cards, Bot Framework, Power Automate

Agentic AI Systems: LangGraph, LangChain, CrewAI, Multi-Agent Orchestration, RAG, Vector Databases (ChromaDB), Local LLMs (Ollama)

Generative AI & LLM Evaluation: LLM Validation (GPT-4, Claude, Gemini), Prompt Engineering, LLM-as-Judge, DeepEval, Model Output Evaluation

API & Service Validation: REST, SOAP, Postman, SOAPUI, Contract Testing, Apigee API Gateway, Data Integrity Checks

CI/CD & DevOps: Azure DevOps Pipelines, Jenkins, GitHub Actions, Git, Automated Quality Gates, Parallel Execution

Cloud Platforms: AWS (Cloud Practitioner), Microsoft Azure (CI/CD, OpenAI, Bot Services), Docker

Programming & Practices: Python, TypeScript/Node.js, React, Microservices Validation, Agile/Scrum, Defect RCA, JIRA, MongoDB

PROFESSIONAL EXPERIENCE

Assistant Manager - AI Automation QA

VOIS (Vodafone Intelligent Solutions) | Sep 2025 - Present

- Architected intelligent automation frameworks using Python, PyTest, Playwright, and BDD, improving maintainability by 40%.
- Deployed Azure OpenAI and Copilot Studio agents for automated test generation, defect prediction, and documentation-driven test case creation.
- Developed scalable validation suites for UI, API, and backend systems, increasing regression coverage by 55%.
- Automated REST and SOAP service validation using Postman and SOAPUI, reducing production defects by 30%.
- Integrated automated quality checks into Azure DevOps and Jenkins pipelines, cutting release validation time by 45%.
- Evaluated LLM-powered features by validating prompt behavior, response accuracy, and conversational consistency.
- Led GenAI proof-of-concepts for AI-assisted test generation, intelligent defect prediction, and Copilot Studio agent development.
- Optimized execution through parallelization and analytics-driven reporting, improving pipeline efficiency by 35%.
- Collaborated with engineering teams to reduce defect leakage by 25% through structured root-cause analysis.

Senior Executive - Automation QA Engineer

VOIS (Vodafone Intelligent Solutions) | Aug 2023 - Sep 2025

- Designed and maintained modular Python-based automation frameworks using PyTest and BDD.
- Automated UI and service-level validation for microservices-based applications.
- Embedded validation workflows into Jenkins CI pipelines to enable continuous quality checks.
- Executed functional, regression, and integration testing in Agile environments.
- Began early exploration of LLM-powered test generation tooling using OpenAI APIs.

Pioneer Technical Graduate - QA Automation

VOIS (Vodafone Intelligent Solutions) | Aug 2022 - Aug 2023

- Built automated verification scripts using Python, PyTest, and BDD frameworks.
- Supported regression testing and API-level validation initiatives.
- Enabled CI-based execution for faster feedback cycles.

AI AGENT PROJECTS (AZURE OPENAI & COPILOT STUDIO)

Copilot Studio Test-Case Automation Agent

Azure OpenAI, Copilot Studio, Power Automate, PyTest, Azure DevOps

- Built an agent that auto-generates PyTest/BDD test cases from natural-language requirements and user stories, integrating with Azure DevOps work items via Power Automate to pull acceptance criteria, scaffold tests, and commit to the repository.
- Reduced manual test-authoring effort by approximately 60%; delivered an adaptive-card chat interface in Microsoft Teams for conversational refinement of generated cases.

MongoDB Query Agent

Azure OpenAI (GPT-4o), Copilot Studio, MongoDB Atlas, Custom Connectors

- Built a natural-language-to-MongoDB-query agent that translates plain-English questions into optimized aggregation pipelines and find() queries, using schema-aware prompt engineering for nested documents and arrays.
- Exposed the agent as a Copilot Studio topic, enabling non-technical stakeholders to query production databases safely through an approval workflow.

Apigee API Creation Agent

Azure OpenAI, Copilot Studio, Google Apigee, OpenAPI 3.0, Power Automate

- Built a conversational agent that scaffolds full Apigee API proxies (OpenAPI spec, proxy bundle, policies) from a plain-English description, cutting API onboarding time from days to under 2 hours.
- Pushed ready-to-deploy proxy bundles to Apigee via the REST management API and integrated with Azure DevOps to trigger automated contract tests immediately after creation.

Documentation-to-Test-Case Generation Agent

Azure OpenAI, Copilot Studio, SharePoint, Playwright, Confluence REST API

- Built an agent that ingests technical documentation (PDF, Word, Confluence, SharePoint) and extracts structured test scenarios (happy paths, edge cases, negative flows) using RAG over an internal corpus.
- Generated Gherkin feature files and Playwright automation scripts directly from documentation, achieving 45% faster test-suite bootstrapping for new projects.

WebMethods Integration Assistant Agent

Azure OpenAI, Copilot Studio, Software AG webMethods, REST/SOAP APIs

- Built a conversational assistant that helps integration developers query webMethods IS service metadata, troubleshoot flow errors, and generate integration stubs.
- Parsed webMethods audit logs and error journals to provide root-cause-analysis suggestions in natural language, cutting incident resolution time by 35%.

INDEPENDENT OPEN-SOURCE AI ENGINEERING PROJECTS

Self-initiated platforms exploring agentic, self-healing quality engineering, built and published independently on GitHub.

Copilot for Playwright - AI QA Console

React, TypeScript, Node.js, Express, Anthropic Claude, Playwright | github.com/upadrastaharshavardhan/Copilot-for-Playwright-v2.1

- Built a full-stack AI console for Playwright covering requirement-to-test generation, Excel/CSV test-pack ingestion generating 5,000-10,000+ data-driven cases in under a second, self-healing locators, failure RCA, and a traceability matrix.
- Designed a dual-mode architecture: a deterministic Smart Template mode (zero cost, no API key) as the default, plus an optional Live AI mode (Claude) with graceful fallback for reliability and cost control.

AgenticTestForge - Multi-Agent Test Generation & Quality Gate

LangChain, Ollama, CrewAI, PyTest, DeepEval | github.com/upadrastaharshavardhan/AgenticTestForge-V2

- Built a 5-agent orchestration pipeline (Requirements Analyst to Test Generator to Quality Evaluator to Refiner to Reporter) converting natural-language requirements into Gherkin plus PyTest BDD suites.
- Implemented an LLM-as-judge quality gate scoring coverage, assertion strength, and hallucination risk, with iterative refinement until a 0.75+ threshold; runs fully local on Ollama at zero token cost.

LocatorForge (XPath Hunter Pro V3) - Locator Intelligence Platform

TypeScript, Playwright, Chrome Extension (Manifest V3) | github.com/upadrastaharshavardhan/XPath-Hunter-Pro-V3---LocatorForge

- Built a locator intelligence engine that scans the DOM, evaluates 9 strategies (Test ID, ARIA role, label, CSS, XPath, and more), and scores candidates on confidence and stability scales.
- Developed a runtime self-healing layer recovering broken locators with confidence-gated auto-heal/warn/review policies, plus a Chrome extension, dashboard, and CLI.

Q-GATE AI - LangGraph CI/CD Quality Gate

LangGraph, Python 3.12, SQLite, Bandit | github.com/upadrastaharshavardhan/qgate-ai

- Built a deterministic PASS/WARN/BLOCK quality gate that analyzes real git diffs, runs parallel security/dependency/regression-risk agents, and selects and executes only impacted tests.
- Added a historical memory layer (SQLite) tracking hotspots and prior failures, with ready-made pipeline templates for GitHub Actions, GitLab CI, Azure DevOps, and Jenkins.

AI Root Cause Investigator (RCI)

LangGraph, NetworkX, ChromaDB, FastAPI | github.com/upadrastaharshavardhan/ai-root-cause-investigator

- Built an evidence-driven RCA system correlating Azure DevOps pipelines, Git history, and Playwright traces/logs with past failures into an impact graph, diagnosing failed runs in under 2 minutes.
- Designed a plugin-based collector/analyzer architecture with a human-in-the-loop approval gate and a ChromaDB memory layer for continuous learning.

TestDNA-AI (Genesis v1) - AI Engineering Memory Platform

FastAPI, Next.js, Knowledge Graph, Semantic Search | github.com/upadrastaharshavardhan/TestDNA-AI-Genesis-v1

- Designed an end-to-end platform (repository scanner to DNA engine to knowledge graph to semantic search to AI copilot) to eliminate duplicate test automation across organizational repositories.
- Authored a 20-document architecture set covering system design, security, deployment, observability, and a plugin SDK for enterprise adoption.

EDUCATION

M.Tech - Computer Science & Engineering (AI & ML) | Jawaharlal Nehru Technological University, Kakinada | 2025

B.Tech - Computer Science & Engineering | VSM College of Engineering | 2018 - 2022

CERTIFICATIONS & AWARDS

- AWS Well-Architected Proficient - Amazon Web Services (Oct 2025)
- Automation Pro I, II, III - Workato (2025)
- Building AI Agents with MongoDB - MongoDB (Apr 2026)
- Memory for AI Applications with MongoDB - MongoDB (Apr 2026)
- Model Context Protocol: Advanced Topics - Anthropic (Mar 2026)
- Claude Code in Action - Anthropic (Mar 2026)
- Generative AI Foundations (Final Exam) - Skillsoft / MIT Horizon (Jan 2025)
- MIT Horizon Critical Emerging Technologies - MIT Horizon (Jan 2025)
- AI Tools Workshop - Be10x (Oct 2025)
- N8N Crash Course - Automate Your Life with AI - LetsUpgrade (Aug 2025)
- DevSecOps Foundation - Vodafone (2023)
- Cyber Security Essentials - Vodafone (2023)
- AWS Cloud Practitioner modules (Migration, Database, Storage, Deploying) - Skillsoft (2024)
- STAR Award - VOIS (Multiple Times), recognized for outstanding contributions to AI-driven quality engineering initiatives